The sixth Schur number is at least 537

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Abstract

The lower bound $S(6) \geq 536$ for the sixth Schur number, found by Fredricksen and Sweet in 2000, has stood for twenty-six years. We break it: we give a partition of $\{1, \ldots, 537\}$ into six sum-free sets, so $S(6) \geq 537$. Through known recursive constructions this also improves the best lower bounds for $S(8)$, $S(11)$, $S(13)$ and for the six-colour Ramsey number $R_6(3)$. Our search produced 131 such partitions, no two of which are related by renaming colours or by a multiplication $x \mapsto rx$ modulo 538. They fall into five families, which we describe. The full list and verification code are supplied in the companion repository.

1 Introduction

A set C of integers is *sum-free* if there are no $x, y \in C$, not necessarily distinct, with $x + y \in C$. The Schur number $S(k)$ is the largest n for which $\{1, \ldots, n\}$ can be partitioned into k sum-free sets. The values $S(1), \ldots, S(5) = 1, 4, 13, 44, 160$ are known, the last one by a large SAT computation of Heule [8]. For $k = 6$ the record has been $S(6) \geq 536$ since 2000, when Fredricksen and Sweet [7] found it by a search over symmetric sum-free partitions; it has been quoted as the best known bound ever since. We improve it by one.

Theorem 1. $S(6) \geq 537$.

Proof. The following six sets are pairwise disjoint, their union is $\{1, \ldots, 537\}$, and none of them contains x, y and $x + y$. This is a finite check; a script that performs it is supplied as `check.py` in the companion repository.

Color 1 (76 integers).

```
  1   5  12  15  19  21  23  25  29  32  39  43  45  49  95
102 106 108 112 119 122 126 128 130 132 136 139 146 150 152
156 163 166 170 172 174 176 180 358 362 364 366 368 372 375
382 386 388 392 399 402 406 408 410 412 416 419 426 430 432
436 443 489 493 495 499 506 509 513 515 517 519 523 526 533
537
```

Color 2 (76 integers).

```
  2   6   7  17  18  22  27  37  41  42  46  51  66  70  81
 89  90 100 101 105 109 110 113 114 124 125 129 133 134 144
145 149 153 157 158 168 169 192 346 369 370 380 381 385 389
393 394 404 405 409 413 414 424 425 428 429 433 437 438 448
449 457 468 472 487 492 496 497 501 511 516 520 521 531 532
536
```

Color 3 (88 integers).

```
  3   8   9  14  20  24  26  30  31  36  47  53  59  63  65
 69  76  82  86  88  92  98 104 115 120 121 127 131 137 142
143 148 154 159 160 165 171 177 181 182 187 188 193 204 334
345 350 351 356 357 361 367 373 378 379 384 390 395 396 401
407 411 417 418 423 434 440 446 450 452 456 462 469 473 475
479 485 491 502 507 508 512 514 518 524 529 530 535
```

Color 4 (70 integers).

```
  4  10  11  13  16  28  33  34  35  40  55  57  79  93  94
 96  99 111 116 117 118 123 135 138 140 141 147 155 161 162
164 167 179 185 203 335 353 359 371 374 376 377 383 391 397
398 400 403 415 420 421 422 427 439 442 444 445 459 481 483
498 503 504 505 510 522 525 527 528 534
```

Color 5 (76 integers).

```
 38  44  48  50  52  54  56  58  60  61  62  64  67  68  71
 72  73  74  75  77  78  80  83  84  85  87  91  97 103 107
173 189 196 197 201 209 212 236 302 326 329 337 341 342 349
365 431 435 441 447 451 453 454 455 458 460 461 463 464 465
466 467 470 471 474 476 477 478 480 482 484 486 488 490 494
500
```

Color 6 (151 integers).

```
151 175 178 183 184 186 190 191 194 195 198 199 200 202 205
206 207 208 210 211 213 214 215 216 217 218 219 220 221 222
223 224 225 226 227 228 229 230 231 232 233 234 235 237 238
239 240 241 242 243 244 245 246 247 248 249 250 251 252 253
254 255 256 257 258 259 260 261 262 263 264 265 266 267 268
269 270 271 272 273 274 275 276 277 278 279 280 281 282 283
284 285 286 287 288 289 290 291 292 293 294 295 296 297 298
299 300 301 303 304 305 306 307 308 309 310 311 312 313 314
315 316 317 318 319 320 321 322 323 324 325 327 328 330 331
332 333 336 338 339 340 343 344 347 348 352 354 355 360 363
387
```

□

Theorem 1 needs only this one partition. Section 2 lists what it implies for other Schur and Ramsey numbers. In Section 3 we describe all 131 partitions of $\{1, \ldots, 537\}$ into six sum-free sets that our search produced. Section 4 explains briefly how the partitions were found. The complete results, code, and supplementary tables are available at `https://github.com/jeffchen006/Schur-Number-6`.

Throughout, $N = 538$, $\mathrm{f}(x) = \min(x, N - x)$, and a colouring c of $\{1, \ldots, 537\}$ is *palindromic* if $c(x) = c(N - x)$ for all x. We number the colours $1, \ldots, 6$ in the order of their least elements.

2 Consequences for other bounds

Lower bounds for larger Schur numbers are built from smaller ones by recursive constructions. Starting from Schur's $S(k+1) \geq 3S(k) + 1$ [1] and the Abbott–Hanson bound $S(k+2) \geq 9S(k)+4$ [2], the strongest known are template constructions: Rowley's $S(k+3) \geq 33S(k)+6$ [3], the bounds $S(k+4) \geq 111S(k) + 43$ and $S(k+5) \geq 380S(k) + 148$ of Ageron et al. [4], and $S(k+2) \geq 10S(k) + 2$ of Bengone et al. [5]. Each of them turns any partition of $\{1, \ldots, p\}$ into k sum-free sets into a larger one, for p as large as the ones considered here. A partition of $\{1, \ldots, n\}$ into k sum-free sets also gives a colouring of the edges of the complete graph on

$\{0,\dots,n\}$ in k colours without a monochromatic triangle (colour the edge ij by the class of $|i-j|$), so $R_k(3) \ge S(k)+2$.

Corollary 2.

$$S(8) \ge 5372, \qquad S(11) \ge 204\,208, \qquad S(13) \ge 2\,042\,082, \qquad R_6(3) \ge 539.$$

The previous bounds were $S(8) \ge 5362$ *and* $S(13) \ge 2\,038\,282$ *[5],* $S(11) \ge 203\,828$ *[4], and* $R_6(3) \ge 538$ *[7, 6].*

Proof. $S(8) \ge 10 \cdot 537 + 2$, $S(11) \ge 380 \cdot 537 + 148$, $S(13) \ge 10 \cdot S(11) + 2 \ge 10 \cdot 204\,208 + 2$, and $R_6(3) \ge S(6) + 2$. The other recurrences above give smaller values for these four numbers. For $S(8)$ we also built the partition explicitly: apply the width-10 template of [5, Table II] to the partition of Theorem 1, with colours 7 and 8 for the labels A and B. The companion script `check.py` builds this partition of $\{1,\dots,5372\}$ into eight sets and checks that all of them are sum-free. □

Bounds for larger k that are derived through these recurrences change in the same way; we list only the four cases in which we checked the previously published values.

3 The 537-partitions and their families

The example above is palindromic, and each of its classes is even sum-free modulo 538. The least elements of its classes are $1, 2, 3, 4, 38, 151$, and the sixth class contains the whole interval $[237, 301]$. These features are captured by the following family of colourings.

Definition 3. A colouring $c\colon \{1,\dots,537\} \to \{1,\dots,6\}$ whose classes are sum-free modulo 538 is in *frame* g if, with $Q = 151$ and $d = 38$,

$$\begin{aligned}
\mathrm{f}(x) < j &\;\Rightarrow\; c(x) \ne j \quad (j = 1,2,3,4),\\
\mathrm{f}(x) < d \text{ or } Q - d < \mathrm{f}(x) < Q &\;\Rightarrow\; c(x) \ne 5,\\
\mathrm{f}(x) \in \{d,\, N - 2Q\} &\;\Rightarrow\; c(x) = 5,\\
\mathrm{f}(x) < Q &\;\Rightarrow\; c(x) \ne 6,\\
\mathrm{f}(x) = Q \text{ or } \mathrm{f}(x) > N - 2Q &\;\Rightarrow\; c(x) = 6.
\end{aligned}$$

In words: the sixth colour does not occur below 151 and fills the middle of the interval, and the fifth colour does not occur below 38 or just below 151. The first class of the example has a simple description as well. Put

$$\begin{aligned}
A &= \{1, 5, 12, 15, 19, 21, 23, 25, 29, 32, 39, 43, 45, 49\},\\
B &= A \cup (151 - A) \cup \{151 + a : a \in A,\ a \le 29\} \cup \{95\},
\end{aligned}$$

a set of 38 numbers, and $C^* = \{x \in \{1,\dots,537\} : \mathrm{f}(x) \in B\}$, a set of 76 numbers. The first class of the example is exactly C^*.

Proposition 4. *The* 131 *partitions of* $\{1,\dots,537\}$ *into six sum-free sets listed in the companion file* `solutions537.txt` *are pairwise distinct, every class of each of them is sum-free modulo* 538*, and no one of them is obtained from another by a map* $x \mapsto rx \bmod 538$ *with* $\gcd(r, 538) = 1$ *followed by renaming the colours. They form the following five families.*

$\mathcal{F}_1$ **(107 partitions).** *Palindromic, in frame* g*, with first class equal to* C^**. The example of Theorem 1 is* $\mathcal{F}_1.001$*.*

$\mathcal{F}_2$ **(14 partitions).** *Palindromic and in frame* g*; the first class differs from* C^* *in one pair* $\{x, 538 - x\}$*.*

$\mathcal{F}_3$ **(5 partitions).** *In frame g with first class C^*, but not palindromic: between* 2 *and* 12 *pairs* $\{x, 538 - x\}$ *receive different colours.*

$\mathcal{F}_4$ **(4 partitions).** *Palindromic, and they satisfy every condition of frame g except the last one only in part: between* 1 *and* 13 *values* $\mathrm{f}(x) > 236$ *do not receive colour* 6*. The first class differs from C^* in between* 2 *and* 8 *numbers.*

$\mathcal{F}_5$ **(1 partition).** *Its classes start at* $1, 2, 3, 4, 7, 16$*. After the substitution* $x \mapsto 43x \bmod 538$ *its classes start at* $1, 2, 3, 4, 38, 151$*, it fails palindromy in* 5 *pairs, and its first class differs from C^* in* 3 *numbers.*

Each partition in $\mathcal{F}_1$–$\mathcal{F}_4$ agrees with the example in at least 462 *of the* 537 *positions.*

Proof. All statements are finite checks, carried out by the companion script `check.py`. □

So the partitions we found are variations of a single solution. Three further observations, from SAT runs without proof certificates, point the same way: (i) with $Q = 151, d = 38$ replaced by any $140 \le Q \le 160$, $39 \le d \le 45$, the conditions of Definition 3 have no solution; (ii) for $d = 38$ and $148 \le Q \le 154$, every solution has a class containing $A \cup (Q - A)$; (iii) of 9,160 changes of one to three colours at positions 5–30 of the 32 partitions known at the time, only two still extend to a palindromic 537-partition.

4 How to reproduce the search

Encoding. For a palindromic colouring use variables $v_{y,k}$ for $y \in \{1, \dots, 269\}$ and $k \in \{1, \dots, 6\}$, meaning "$\mathrm{f}(x) = y$ has colour k". The clauses are: $v_{y,1} \vee \cdots \vee v_{y,6}$ for each y; the clause $\neg v_{\mathrm{f}(x),k} \vee \neg v_{\mathrm{f}(y),k} \vee \neg v_{\mathrm{f}(x+y),k}$ for all $1 \le x \le y$, $x + y \le 537$ and all k; and one unit clause for each condition of Definition 3. (A palindromic colouring with sum-free classes is automatically sum-free modulo 538.) The script `gen.py` in the companion repository writes this CNF and decodes a model.

Decoder caveat. The supplied historical generator does not impose at-most-one-colour clauses. Its decoder selects the lowest-numbered true colour at each position. This preserves sum-freeness, but can discard a colour required by a positive unit clause, so the output need not satisfy frame g. For example, a satisfying model can decode positions 237 and 301 as colour 1, although the frame requires colour 6 there. This does not affect the independently checked certificates reported here. To guarantee the frame after decoding, add $\neg v_{y,i} \vee \neg v_{y,j}$ for each position y and $1 \le i < j \le 6$, or choose a required colour first and check all frame conditions after decoding. The repository includes a script that reproduces this caveat.

Solving. With Kissat 4.0.4 [9]:

```
python3 gen.py cnf > g.cnf
kissat --seed=1 g.cnf > g.out
python3 gen.py decode g.out
```

On one core this took 286 s with seed 1 and 4 s with seed 2; both solutions lie in $\mathcal{F}_1$, and different seeds give different members.

The other families. For $\mathcal{F}_3$ drop palindromy: use one variable per $x \in \{1, \dots, 537\}$, forbid monochromatic $x + y \equiv z \pmod{538}$, and impose the frame conditions on $\mathrm{f}(x)$ (`gen.py -nopal`; 41 s with seed 2, while seed 1 did not finish in 15 minutes). For $\mathcal{F}_4$ drop the condition that colour 6 fills $\mathrm{f}(x) > 236$ (`gen.py -nomid`); this allows $\mathcal{F}_4$, but a given run may still return a member of $\mathcal{F}_1$. $\mathcal{F}_5$ came from a list-colouring run: let c be the example and $c_m(x) = c(mx \bmod 538)$ for

$m = 3, 25, 221$, with colours renamed to agree with c as much as possible; each x may only take one of the colours $c_3(x), c_{25}(x), c_{221}(x)$, and no other condition is imposed.

Where frame g came from. An evolutionary search over depth-first search programs, in the spirit of FunSearch [10], produced valid colourings of $\{1, \ldots, n\}$ for n up to 528 from scratch, without seed colourings. Those programs restrict the search by rules that ban a colour below a given value of f, and the rules used by the best colourings had the shape of Definition 3, with Q and d as parameters. At $n = 537$ a sweep over $140 \le Q \le 160$ found no solution for $39 \le d \le 45$ and the first one at $(Q, d) = (151, 38)$.

The work was done by a group of AI agents (Claude Code sessions, with OpenAI Codex for second opinions) on a 192-core server over about a day and a half; every result was recorded with a command that reproduces it and checked by a second agent.

Data and code. The companion repository contains the full list of 131 partitions, a separate PDF table, the example certificate, the 5372-element eight-colouring, `check.py`, and `gen.py`. To check the results, run `python3 check.py` from the repository root with Python 3 and its standard library. The paper retains one complete partition; the remaining partitions and code are supplementary material.

References

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